

PH101: 2023

Special Theory of Relativity

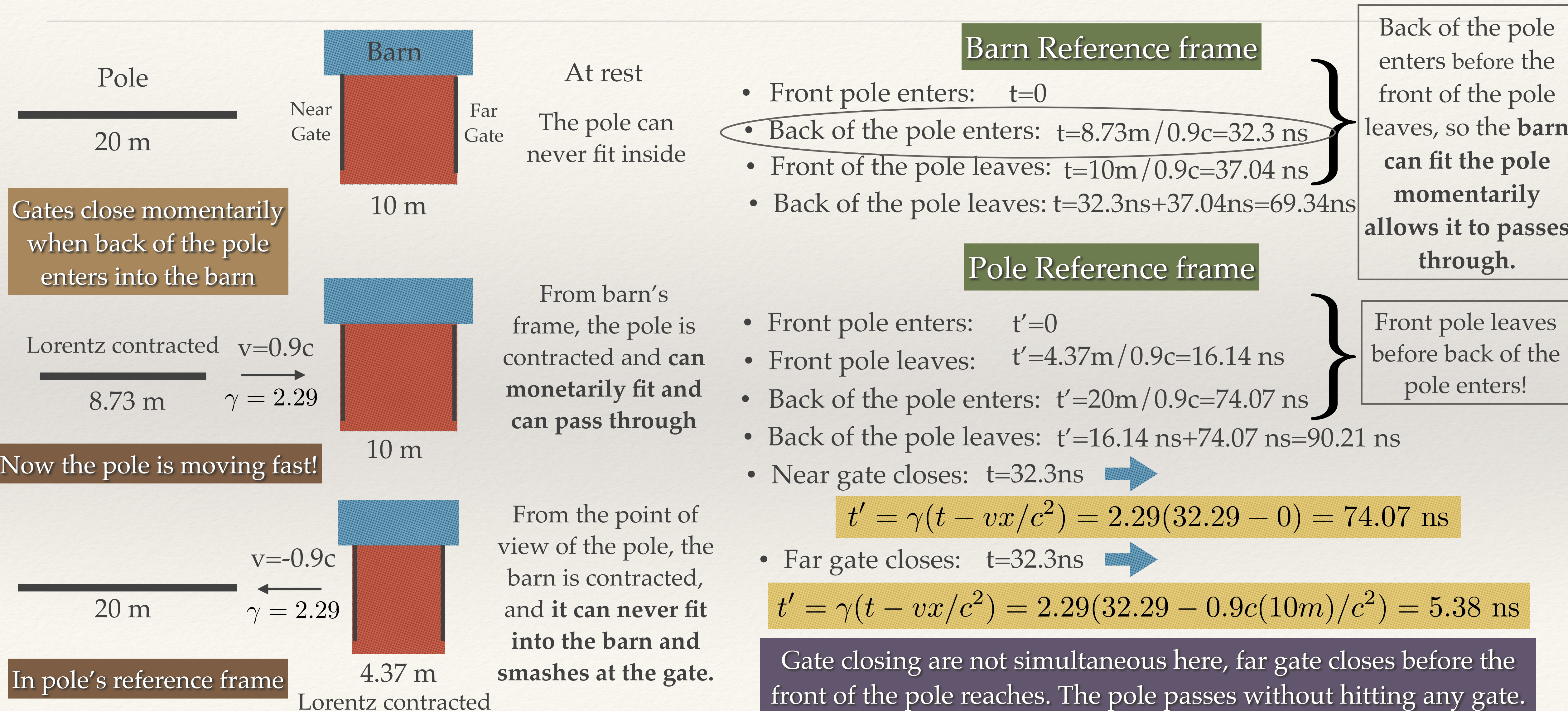
Lecture-3

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Recap

- Non-invariance of Maxwell's equations under Galilean Transformation
 - Michelson Morley Experiment and null result
 - Absence of ether and constancy of velocity of light in any inertial frame
 - Einstein's postulates of Special Theory of Relativity (STR)
 - Lorentz Transformation to relate space time coordinates of inertial frames abiding by the postulates of STR
 - The main consequences of Lorentz transformations are:
 - Length contraction: $l = \frac{l_0}{\gamma}$
 - Time dilation: $T = \gamma\tau$
- $\gamma = \frac{1}{\sqrt{1 - v^2/c^2}} > 1$ Lorentz factor

The Pole Barn Paradox



Relativistic transformation of Velocity

Consider, that a particle is moving with velocity (u_x, u_y, u_z) in the S frame. We have to find (u'_x, u'_y, u'_z) in the S' frame moving with velocity v along x wrto S .

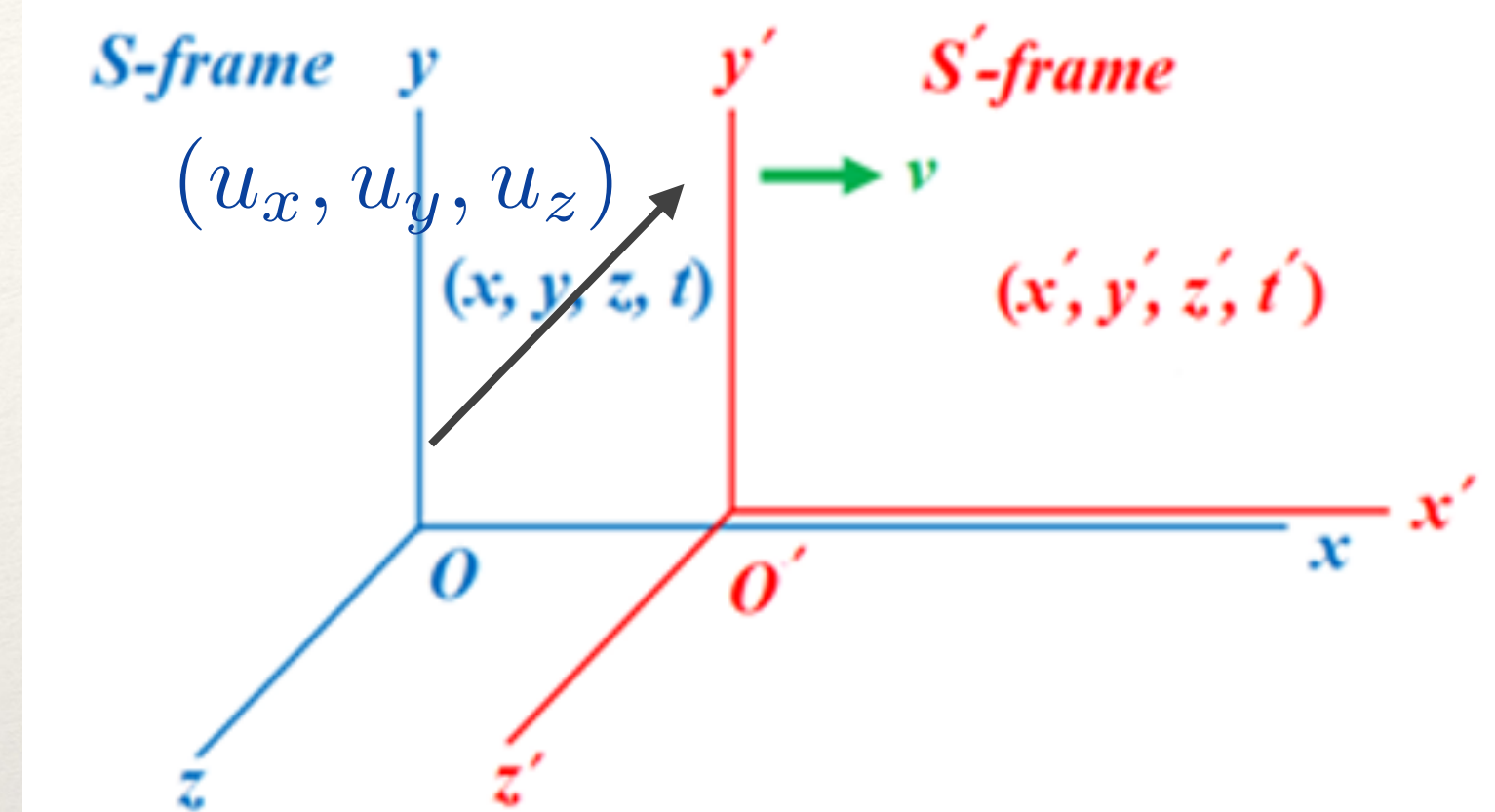
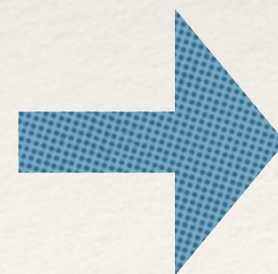
Lorentz transformation

$$x' = \gamma(x - vt), \quad y = y', \quad z = z', \quad \text{and} \quad t' = \gamma\left(t - \frac{v}{c^2}x\right)$$

The components of the velocity are

$$u'_x = \frac{dx'}{dt'} = \gamma\left(\frac{dx}{dt} - v\frac{dt}{dt'}\right), \quad u'_y = \frac{dy'}{dt'}, \quad u'_z = \frac{dz'}{dt'}, \quad \text{and} \quad dt' = \gamma\left(dt - \frac{v}{c^2}dx\right)$$

A little algebra:



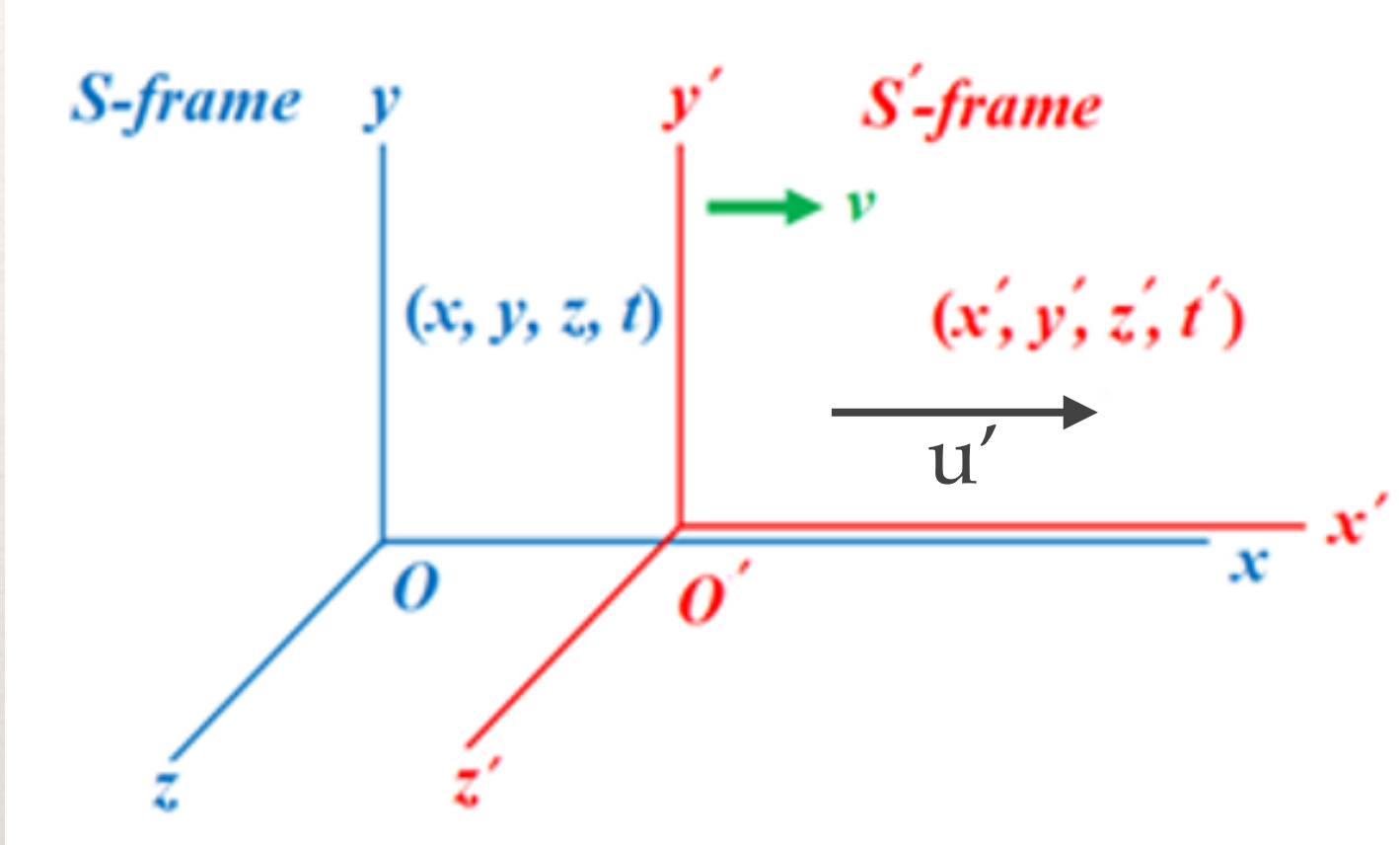
Hence we get the following relations,

$$\begin{aligned} u'_x &= \frac{u_x - v}{1 - vu_x/c^2} \\ u'_y &= \frac{u_y}{\gamma[1 - vu_x/c^2]} \\ u'_z &= \frac{u_z}{\gamma[1 - vu_x/c^2]} \end{aligned}$$

Just changing the sign v to $-v$, we get the inverse transformation.

Velocity transformation/Addition of velocities

Longitudinal velocity addition



An object moves at speed u' with respect to frame S' . And frame S' moves at speed v with respect to frame S in the same direction as the motion of the object. What is the speed, u , of the object with respect to frame S ?

Given, $\Delta x' / \Delta t' = u'$

Our goal is to find $u \equiv \Delta x / \Delta t$.

Following inverse Lorentz transformation,

$$\Delta x = \gamma (\Delta x' + v \Delta t')$$

$$\Delta t = \gamma (\Delta t' + v \Delta x' / c^2)$$

$$\gamma \equiv 1 / \sqrt{1 - v^2 / c^2}$$

For a bullet, a train, and a gun,
Adding the speeds can be fun.
Take a trip down the path
Paved with Einstein's new math,
Where a half plus a half isn't one.

$$u \equiv \frac{\Delta x}{\Delta t} = \frac{\Delta x' + v \Delta t'}{\Delta t' + v \Delta x' / c^2}$$

$$= \frac{\Delta x' / \Delta t' + v}{1 + v (\Delta x' / \Delta t') / c^2}$$

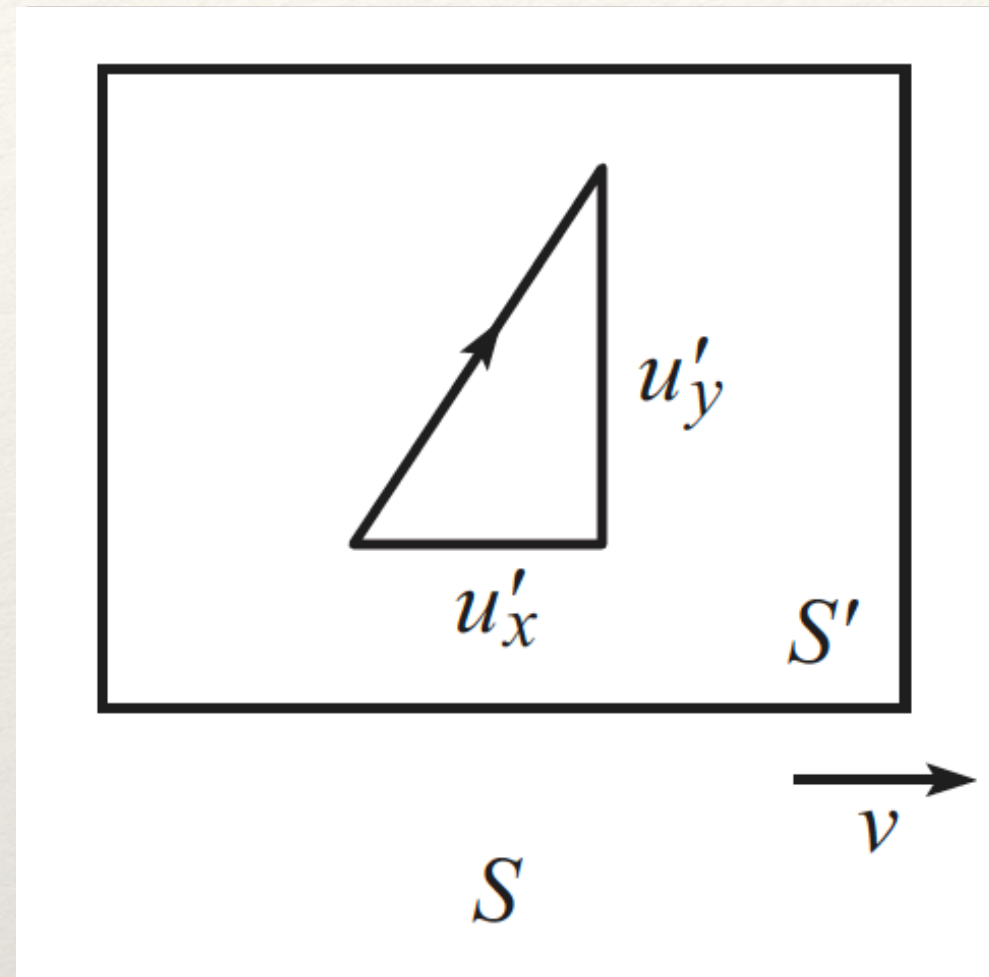
$$= \frac{u' + v}{1 + u' v / c^2}$$

$$u = \frac{u' + v}{1 + u' v / c^2}$$

For $u' v \ll c^2$, it reduces to $u \approx u' + v$

If $u' = c$ or $v = c$, then we find $u = c$,

Transverse Velocity Addition



Consider the following general two-dimensional situation. An object moves with velocity (u'_x, u'_y) with respect to frame S' . And frame S' moves with speed v with respect to frame S , in the x direction. What is the velocity, (u_x, u_y) , of the object with respect to frame S ?

Along x-axis we can use the velocity of addition as derived.

$$u_x = \frac{u'_x + v}{1 + u'_x v / c^2}$$

Question: What is u_y ?

$$u_y \equiv \Delta y / \Delta t$$

$$\Delta y' / \Delta t' = u'_y$$

Lorentz
transformation

$$\Delta y = \Delta y'$$

$$\Delta t = \gamma(\Delta t' + v\Delta x' / c^2)$$

$$u_y \equiv \frac{\Delta y}{\Delta t} = \frac{\Delta y'}{\gamma(\Delta t' + v\Delta x' / c^2)} = \frac{\Delta y' / \Delta t'}{\gamma(1 + v(\Delta x' / \Delta t') / c^2)} = \frac{u'_y}{\gamma(1 + u'_x v / c^2)}.$$

Examples

Example: An object moves with a speed c in the rest frame along x' axis. Calculate its speed in the lab frame. Rest frame moves with speed v relative to the lab frame also along x' axis.

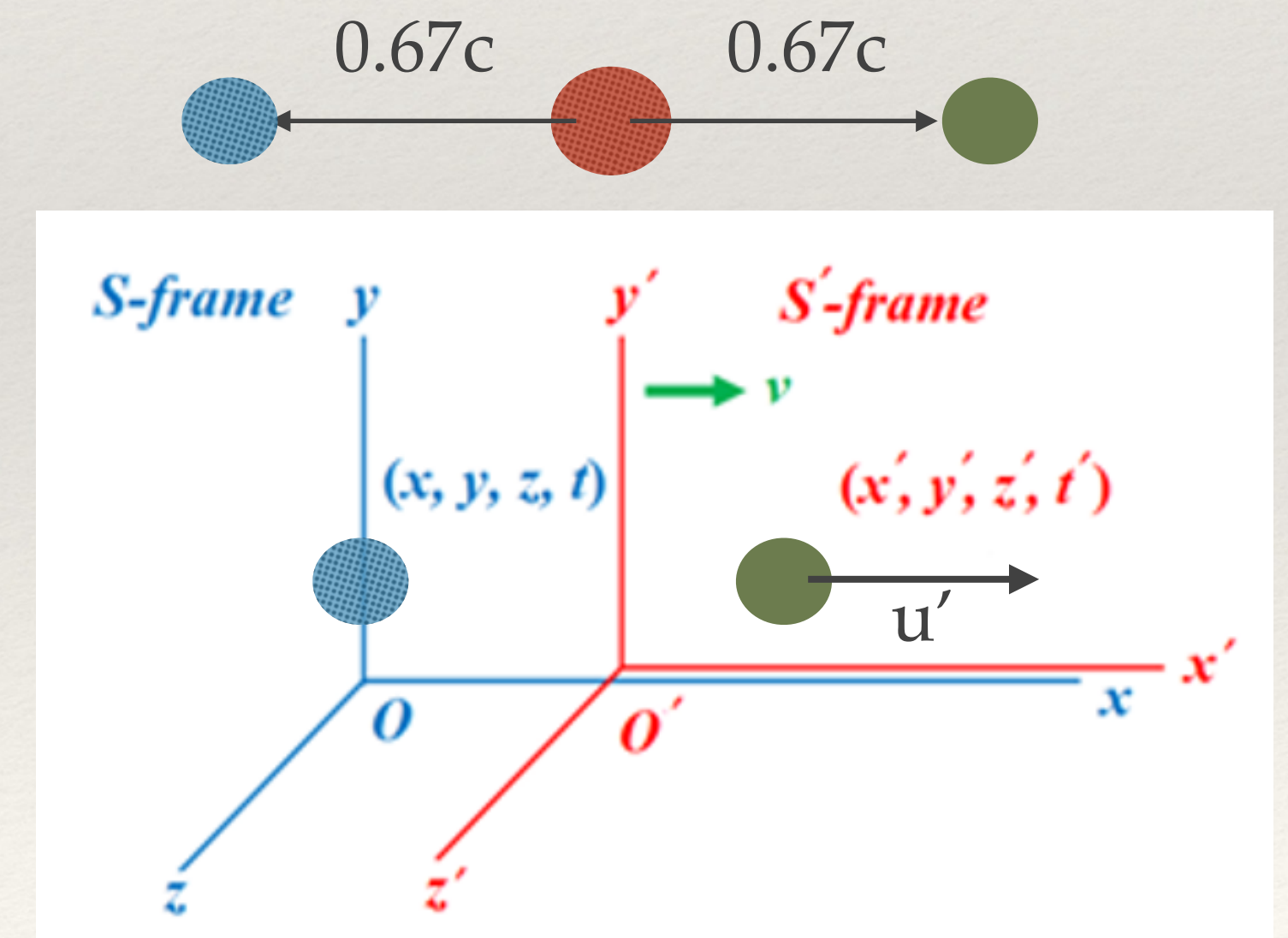
$$u = \frac{u' + v}{1 + u'v/c^2} = \frac{c + v}{1 + cv/c^2} = \frac{c + v}{c(c + v)} c^2 = c.$$

Example: Two electrons leave a radioactive sample in opposite directions, each having a speed $0.67c$ with respect to the sample. Calculate the speed of one electron relative to the other relativistically.

and

$$u' = 0.67c \quad v = 0.67c$$
$$u = \frac{u' + v}{1 + u'v/c^2} = \frac{(0.67 + 0.67)c}{1 + (0.67)^2} = \frac{1.34}{1.45} c = 0.92c.$$

The relativistic addition of any velocity to the velocity of light merely gives again the velocity of light. The relativistic addition of two velocities, each less than c , is itself a velocity less than c .



Example

Example: The velocity of light in matter is less than c . The index of refraction, n , is used to specify the speed in a medium:

$$n = \frac{c}{\text{velocity of light in the medium}}$$

$n = 1$ corresponds to empty space; in matter $n > 1$. The slowing can be appreciable: for water $n = 1.3$. The problem is to find the speed of light through a moving liquid. For instance, consider a tube filled with water. If the water is at rest, the velocity of light in the water with respect to the laboratory is $u = c/n$. What is the speed of light when the water is flowing with speed v ?

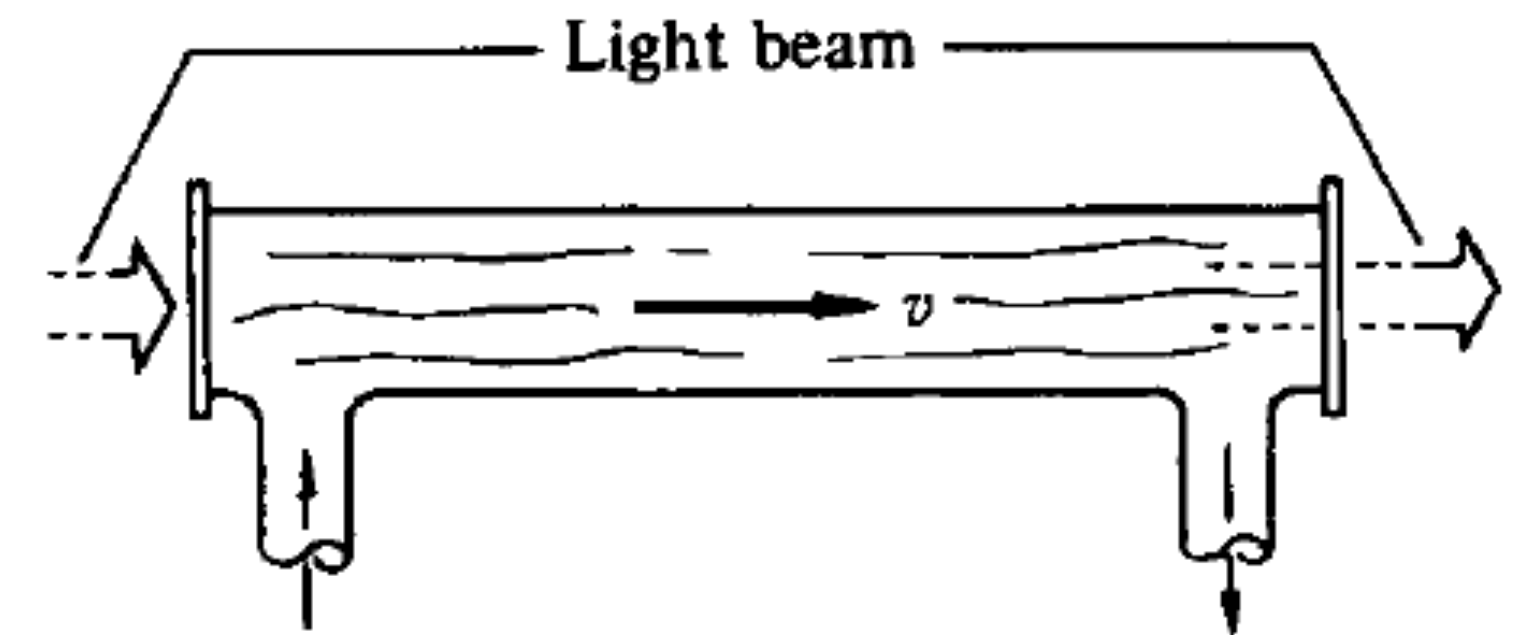
Velocity of light in the **rest frame** is c/n ,

Velocity of light in the **lab frame** is

$$u = \frac{u' + v}{1 + u'v/c^2} = \frac{c/n + v}{1 + v/nc} = \frac{c}{n} \left(\frac{1 + nv/c}{1 + v/nc} \right)$$

Expanding and neglecting the second order term, we get

$$u = \frac{c}{n} \left(1 + \frac{nv}{c} - \frac{v}{nc} \right) = \frac{c}{n} + v \left(1 - \frac{1}{n^2} \right)$$



The light appears to be dragged by the fluid, but not completely. Only the fraction $f = 1 - 1/n^2$ of the fluid velocity is added to the speed of light c/n . This effect was observed experimentally in 1851 by Fizeau.

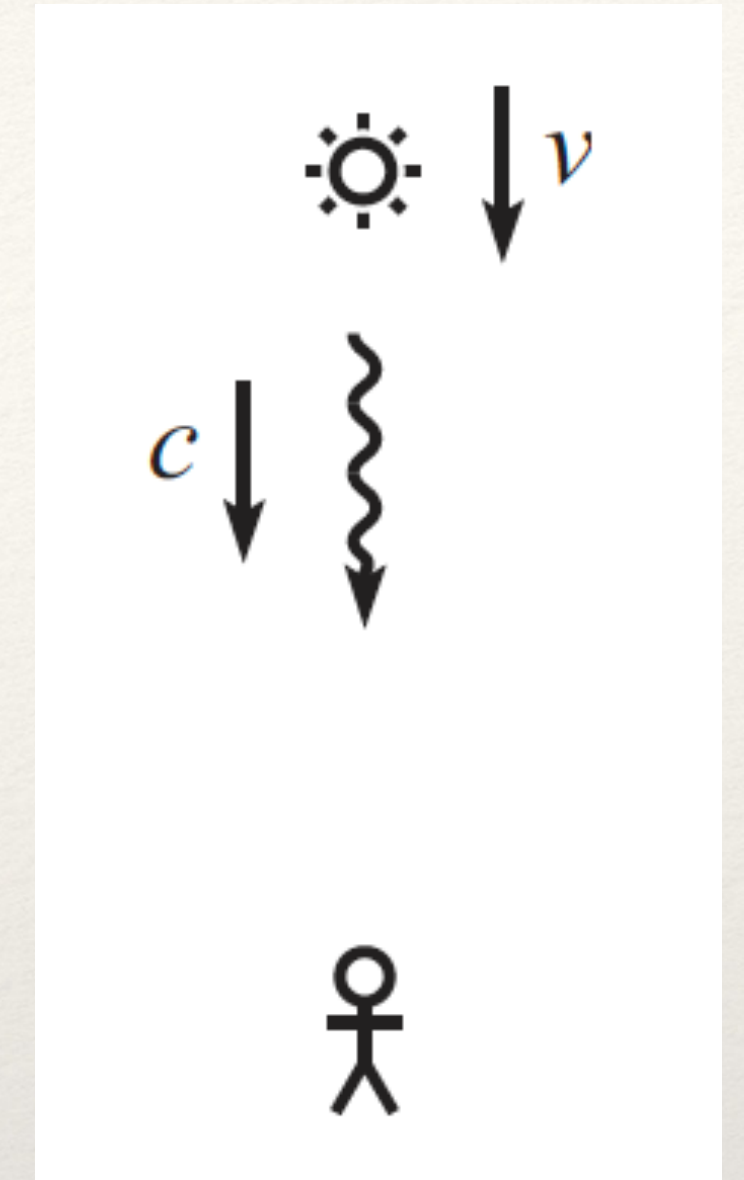
Doppler effect

The Doppler effect is a shift in frequency due to the motion of a source or an observer. The Doppler shift occurs for light as well as for sound.

Applications:

- The motion of distant receding galaxies comes from studies of the Doppler shift of their spectral lines.
- Satellite tracking and radar speed traps.

Q: Consider a source that emits flashes at frequency ν' (in its own frame) while moving directly toward you at speed v . With what frequency do the flashes hit your eye?



The time between emissions in the source's frame is $\Delta t' = 1/\nu'$

The time between emissions in your frame : $\Delta t = \gamma \Delta t'$ **Time dilation**

The distance the photons of one flash have traveled (in your frame) by the time the next flash occurs is $c\Delta t = c\gamma\Delta t'$

During this time the source has travelled: $v\Delta t = v\gamma\Delta t'$

At the instance next flash occurs, the photons are behind you by the distance: $c\Delta t - v\Delta t = (c - v)\gamma\Delta t'$

Doppler effect continued

The time between the arrival of the flashes at your eye

$$\Delta T = \frac{1}{c}(c - v)\gamma\Delta t' = \frac{1 - \beta}{\sqrt{1 - \beta^2}} \Delta t'$$

$$\Delta T = \sqrt{\frac{1 - \beta}{1 + \beta}} \left(\frac{1}{\nu'} \right) \quad \nu = \frac{1}{\Delta T}$$

$$\nu = \sqrt{\frac{1 + \beta}{1 - \beta}} \nu'$$

If the source is moving towards you, then, $\beta > 0$ $\nu > \nu'$

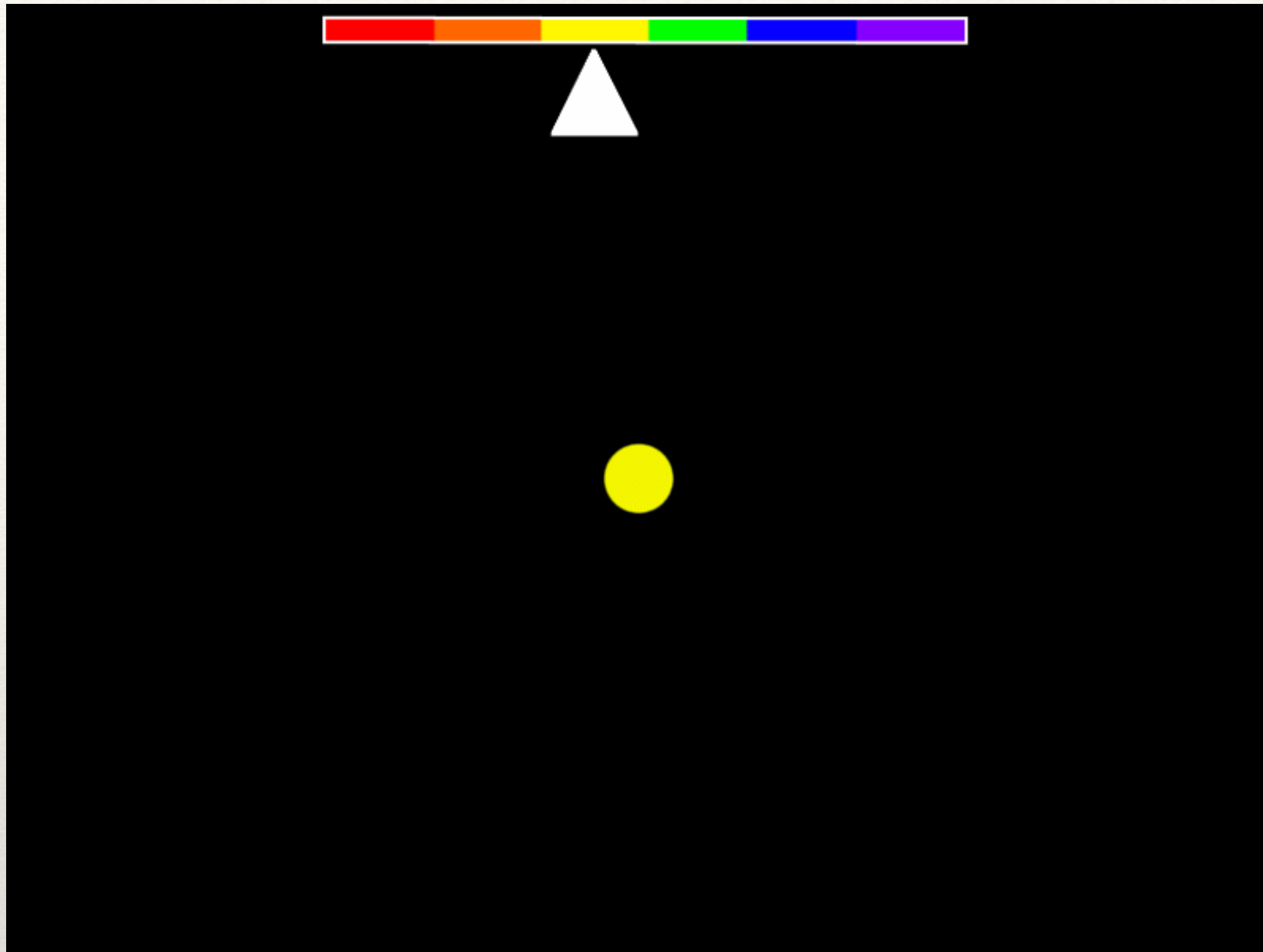
The light is blue shifted

If the source is going away from you, then, $\beta < 0$ $\nu < \nu'$

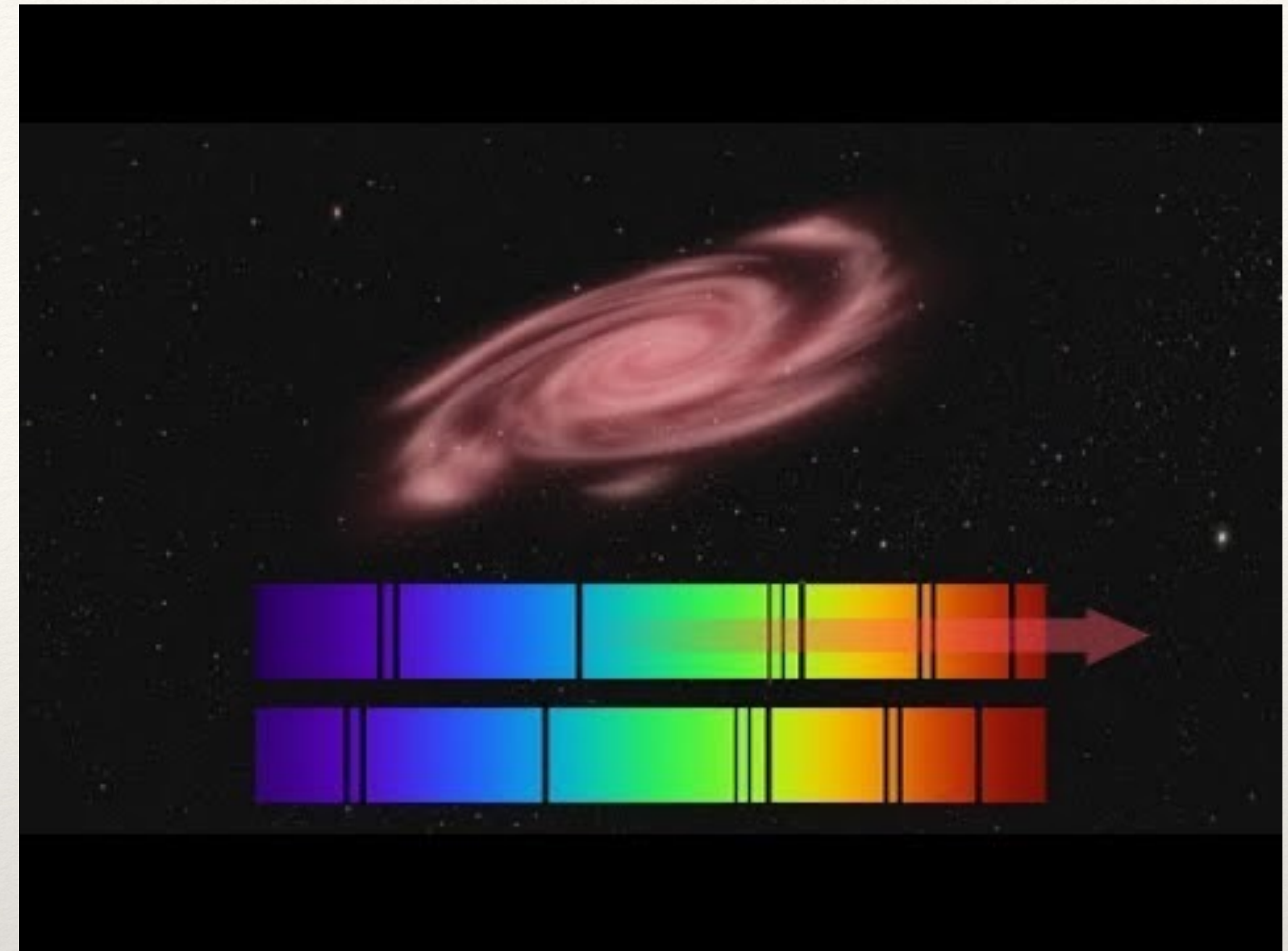
$$\nu = \sqrt{\frac{1 - \beta}{1 + \beta}} \nu'$$

The light is red shifted





Doppler effect: yellow (~575 nm wavelength) ball appears greenish (blueshift to ~565 nm wavelength) approaching observer, turns orange (redshift to ~585 nm wavelength) as it goes away. To observe such a change in color, the object needs to be traveling at approximately 5,200 km/s, or about 75 times faster than the speed record for the fastest man-made space probe: source: Wikipedia.



Around 1929, Edwin Hubble discovered that the Universe is expanding, by observing most other galaxies are moving away from us as the **Light from these galaxies is shifted to longer (and this means redder) wavelengths** - in other words, it is 'red-shifted'.

Summary

- There arise many paradoxes in Special relativity, which essentially questions the notion of simultaneity in different inertial frames, but consistent with Lorentz transformation. Pole Barn paradox is one of them.
- Velocity addition is more non-trivial in STR, due to the presence of Lorentz factor. Reduces to Galilean rule at low velocity. Along the direction of motion,

$$u = \frac{u' + v}{1 + u'v/c^2}$$

$$u' = \frac{u - v}{1 - uv/c^2}$$

- The frequency of a light source also gets shifted due to its motion with respect to observer.

- If the source is moving towards, then it is blue-shifted

$$\nu = \sqrt{\frac{1 + \beta}{1 - \beta}} \nu'$$

- If the source is moving away, then it is red-shifted

$$\nu = \sqrt{\frac{1 - \beta}{1 + \beta}} \nu'$$